

HOW SALESFORCE BUILT ITS PLATFORM BUSINESS

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APRIL 2024 | CISR WP NO. 462 AND MIT SLOAN WP NO. 7364-24 | 31 PAGES

CASE STUDY

an in-depth description of a firm's approach to an IT management issue (intended for MBA and executive education)

PLATFORMS

DESIGNED FOR DIGITAL

ECOSYSTEMS

ENTERPRISE ARCHITECTURE

In 2023, Salesforce was the provider of the CRM system with the biggest market share. Besides providing a CRM software product, the company also offered an external developer platform called Salesforce Platform. Salesforce Platform enabled CRM customers to customize their CRM system and allowed partners (third-party developers) to develop cloud apps that could be distributed via Salesforce's AppExchange marketplace. Since its official launch in 2008, Salesforce Platform had become a major revenue contributor for the company, including from over 7,000 apps in AppExchange.

Building a successful platform business extended far beyond building a technology platform to nurturing an entire ecosystem around it. Based on interviews with twelve executives, this case study illustrates what it takes for a product company to build a platform business—developing and scaling the business model, engineering the technology platform, growing a vivid community of developers, and marketing and selling the platform—and how Salesforce addressed related challenges.

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HOW SALESFORCE BUILT ITS PLATFORM BUSINESS

In 2023, Salesforce was ranked as the number one customer relationship management (CRM) system provider in the world.[1] A key offering for the company was Salesforce Platform, a cloud-based platform enabling the development and deployment of applications for use with Salesforce offerings, with which Salesforce had built an ecosystem of over 4,500 third-party developers who were producing and selling their own apps on top of the platform.

Just as mobile app developers could use Apple's iOS platform to provide their own third-party apps to iPhone users via the company's App Store, Salesforce customers and external developers could use Salesforce Platform to extend the functionality of Salesforce's CRM system or build and offer solutions via the Salesforce AppExchange marketplace.[2] For example, DocuSign offered its eSignature app—which allowed Salesforce customers to "send, sign, track and save agreements in Salesforce" using electronic signatures[3]—via AppExchange. And Veeva Systems, which provided "cloud-based software for the global life sciences industry," built its $2 billion business on top of Salesforce Platform.[4]

1 In April 2023, International Data Corporation (IDC) ranked Salesforce as the #1 CRM provider in its Worldwide Semiannual Software Tracker, the tenth time Salesforce has secured the top spot. See "Salesforce Ranked #1 CRM Provider for 10th Consecutive Year," News & Insights, Salesforce, April 25, 2023, https://www.salesforce.com/news/stories/idc-crm-market-share-ranking-2023/.

2 In fact, advice from Steve Jobs inspired Salesforce to create its marketplace before the iPhone even came out. Salesforce later gifted the App Store trademark to Apple. Amanda Nelson, "How Advice from Steve Jobs Inspired the AppExchange," *The 360 Blog* (blog), Salesforce, August 22, 2022, https://www.salesforce.com/blog/steve-jobs-inspired-appexchange/.

3 DocuSign, Inc. provided products, including its DocuSign eSignature app listed on AppExchange, that allowed organizations to manage agreements electronically. "DocuSign eSignature for Salesforce: The trusted eSignature solution," Salesforce AppExchange, accessed December 7, 2023, https://appexchange.salesforce.com/appxListingDetail?listingId=a0N30000001taX4EAI.

4 Veeva Systems Inc. listed its Veeva CRM app on AppExchange. "Veeva CRM," Salesforce AppExchange, accessed December 7, 2023, https://appexchange.salesforce.com/appxListingDetail?listingId=a0N300000016ZRTEA2.

This case study was prepared by Martin Mocker, Professor at ESB Business School, Reutlingen University, and Academic Research Fellow at the MIT Center for Information Systems Research (CISR); and Ina M. Sebastian, Research Scientist at the MIT Center for Information Systems Research (CISR). The case was written for the purpose of class discussion rather than to illustrate effective or ineffective handling of managerial situations. The authors would like to acknowledge and thank the executives at Salesforce for participating in the case study.

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The AppExchange is the hub of the Salesforce ecosystem. It is a marketplace where customers find solutions that solve business problems, either as add-ons to products Salesforce is selling or to solve unique problems in unique industries for unique user populations that may otherwise have nothing to do with Salesforce.

WOODSON MARTIN, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), APPEXCHANGE, SALESFORCE[5]

Salesforce Platform, which the company introduced in 2004, contributed almost $6 billion (about 20 percent) to Salesforce's total revenue in FY 2023.[6] Salesforce allowed third-party developers—independent software vendors (ISVs) such as DocuSign and Veeva that the company called "partners"—to build their applications using Platform-provided functionality to reduce time-to-market and effort. For example, as part of Platform, Data Cloud[7] provided functionality to unify real-time data from various data sources including customer systems, social media, and IoT sensors into a single customer profile. Developers building apps for AppExchange on top of Platform could utilize Data Cloud's functionality and thus did not have to develop that kind of functionality themselves.

Partners offering an app via AppExchange gained exposure to Salesforce's more than 150,000 CRM customers.[8] In return, Salesforce received a cut of the revenue partners made in selling their offerings via AppExchange.

Salesforce's primary reason for introducing Platform had been to allow any Salesforce customer to customize and extend their own implementation of the Salesforce CRM system to adapt it to their specific needs—without Salesforce having to modify the original CRM system for every customer. To use Platform, customers paid subscription fees to Salesforce ranging from $25 to $100 per user per month.[9]

When customers ask "can you change the software to do this?" I say no, I can't, but I can give you a platform capability that will allow you to do that easily [yourself]. If we wrote it as a hard-coded feature for each customer differently, we would not be able to scale to the number of customers we have today.

PARKER HARRIS, CO-FOUNDER AND CHIEF TECHNOLOGY OFFICER, SALESFORCE

Platform played an important role in the company's growth not just in terms of direct revenue, but also indirectly: Salesforce's own core CRM products were built on Platform. Partner apps helped make those core products more attractive by constantly introducing new, innovative functionality or by tailoring the products for industries not yet served by Salesforce. Also, the company recognized increased customer stickiness resulting from Platform: those customers who had extended their Salesforce CRM system using AppExchange apps were much more likely to renew their CRM subscriptions.

Building a successful platform business, however, was technologically and managerially challenging. Salesforce designed and developed the technology platform, made its functionality available to developers via application programming interfaces (APIs), and created the AppExchange marketplace. The company also had to enable others to use Platform by creating and maintaining documentation, training modules, and certifications on its

5 Woodson Martin left his role at Salesforce in September 2023.

6 Salesforce's annual report for FY 2023 combined reporting for Salesforce Platform and Salesforce's Slack product in a "Platform and Other" segment. Salesforce, *FY23 Annual Report*, March 8, 2023, 6, 58, https://s23.q4cdn.com/574569502/files/doc_financials/2023/ar/salesforce-fy-2023-annual-report.pdf.

7 Salesforce originally introduced Data Cloud as Genie Customer Data Cloud in September 2022, dropping Genie from the name in 2023.

8 "Deliver trusted AI with the world's #1 CRM.," Salesforce, accessed December 7, 2023, https://www.salesforce.com/campaign/worlds-number-one-CRM/.

9 "Platform Pricing," Salesforce, accessed December 7, 2023, https://www.salesforce.com/editions-pricing/platform/. Typical users of Platform licenses were developers and administrators within an organization.

Trailhead learning website.[10] It attracted and supported a developer community. And beyond this, the company marketed and sold Platform and AppExchange apps to customers. In other words, Salesforce's efforts extended far beyond building a technology platform to nurturing an entire ecosystem around it.

In many ways, Platform was different from the other products Salesforce sold. For one, Platform's business had multiple stakeholders with different priorities.

The Platform general manager looks after three constituents. One is our external customers. The second is internal: most of our products, such as Sales Cloud and Service Cloud, are built on Platform. The third is partners that build something onto the platform to put it into the AppExchange marketplace. These are very different constituents or stakeholders, all using the platform in different ways. I often envy [general managers] who just have one stakeholder to worry about.

PATRICK STOKES, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), PLATFORM, SALESFORCE[11]

Dealing with multiple stakeholders required continuously making decisions about how to allocate limited engineering resources. Platform's business leaders could employ engineers to develop either functionality that would yield direct revenue, via Platform and the AppExchange; or indirect revenue, by enabling the company's core products to offer new features based on improved Platform functionality. Leaders had to make strategic decisions, such as whether Platform should stay focused on enabling and enhancing the core CRM product or become a more generalized platform that would compete directly with Amazon's AWS or Microsoft's Azure platforms.

SALESFORCE COMPANY BACKGROUND

Founded in 1999 and publicly listed in 2004, by 2023 Salesforce, Inc. (Salesforce) had grown to become the market leader in CRM systems and the "world's largest provider of enterprise applications."[12] The company had almost 80,000 employees and reported over $31 billion in revenue for its 2023 fiscal year that ended January 31, 2023. See appendix A for the company's financials.

An Overview of Salesforce Offerings

In FY 2023, over 90 percent of Salesforce's revenues came from subscription and support fees for the software products that made up its Customer 360 integrated CRM suite (see appendix B). Platform was an offering available as part of Customer 360 while at the same time enabling it. Salesforce was best known for its cloud-based CRM software-as-a-service (SaaS) solution. Customers paid a monthly subscription fee[13] to use software that Salesforce provided and maintained in the cloud rather than buying and installing such software on premises on their own infrastructure.

Salesforce's core CRM system comprised its Service Cloud, Sales Cloud, Marketing Cloud, and Commerce Cloud offerings, with the Service and Sales clouds generating 25 and 24 percent respectively of FY 2023 subscription and support revenues and the combined Marketing and Commerce clouds contributing 16 percent. These offerings allowed a company's marketing, sales, and service employees to enter and analyze data related to all customer touchpoints. For example, if a company sold a product to a customer, emailed out advertising to a customer, or received a customer complaint about an issue with a product, the events could be recorded in the

10 Salesforce developed Trailhead, a gamified learning platform, to house its skill development programs and training modules. For example, see Trailhead training module "Platform Developer I," Credentials, Salesforce Trailhead, accessed December 7, 2023, https://trailhead.salesforce.com/en/credentials/platformdeveloperi.

11 Patrick Stokes assumed the role of Executive Vice President of Product and Industries Marketing in June 2023.

12 "Salesforce Tops SAP as World's Largest Enterprise Application Company," *The Software Report*, September 20, 2022, https://www.thesoftwarereport.com/salesforce-tops-sap-as-worlds-largest-enterprise-application-company/.

13 For example, in December 2023, Sales Cloud customers paid between $25 and $500 per user per month, depending on the edition they subscribed to; Sales Cloud Enterprise edition, which according to Salesforce was the most popular edition, cost $165 per user per month. "Salesforce Product Pricing," Salesforce, accessed December 8, 2023, https://www.salesforce.com/editions-pricing/overview/.

Salesforce CRM system. In this way, companies could share and analyze data enterprise-wide and make it easier to enforce, track, and automate standardized business processes.

The Beginnings of Salesforce Platform

Started as sforce in 2004 and launched officially as force.com in 2008,[14] Salesforce Platform was a platform-as-a-service (PaaS) offering that allowed developers to build their own SaaS apps by reusing and extending the core functionality underlying Salesforce's CRM products.

Underneath Sales Cloud and Service Cloud is a lot of the same functionality. We said, "What if we could open that up and give those same tools to our customers and let them build whatever applications they need?" The focus on partners grew out of that.

WOODSON MARTIN, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), APPEXCHANGE, SALESFORCE

In Salesforce's first annual report as a public company in 2005, one of five key elements of the company's strategy was to "encourage the development of third-party applications on our […] platform." In September 2005, Salesforce introduced AppExchange as a marketplace to "browse, test-drive, share, and install applications developed on our […] platform."[15] Up until FY 2014, Salesforce noted in its annual reports that the markets for its Salesforce Platform remained relatively new and it was "uncertain whether our efforts will ever result in significant revenue for us." After 2014, however, Platform revenues accelerated rapidly, growing at a faster rate than total company revenues; and in FY 2023, they accounted for about 20 percent of all company revenues.[16]

RUNNING A THRIVING PLATFORM BUSINESS

In 2023, about 2,500 people were working on Platform across the entire Salesforce organization.[17] This included roughly 1,300 engineers who were building the technology behind Platform and an assortment of other teams: product teams defining the business model and platform features, including AppExchange; marketing teams promoting Platform to customers and partners and building stakeholder communities; and sales teams selling Platform to customers.

Managing the Platform Offering

Just as with any other product, Salesforce had to specify and evolve the features of the Platform offering. The company also had to define Platform's business model—how stakeholders would interact with the offering and how it would make money. This was the focus of the Platform product management team, led by the general manager for Platform, who held responsibility for Platform's profit and loss (P&L).

DEVELOPING PLATFORM'S BUSINESS MODEL

Salesforce described Platform's key functionality as enabling "customers, independent software vendors ("ISVs") and third-party developers to create, test and deliver cloud-based apps" that could be "marketed and sold on the AppExchange […] enterprise cloud marketplace or […] directly by software vendors."[18]

14 Michael Arrington, "Salesforce Enters Custom Application Market With Force.com," *TechCrunch*, September 14, 2007, https://techcrunch.com/2007/09/13/salesforce-enters-custom-application-market-with-forcecom/.

15 Salesforce, *Salesforce.com Annual Report 2006*, March 15, 2006, 7.

16 Salesforce, *FY23 Annual Report*, 58.

17 See appendix C for a high-level organization chart of the interviewees for this case study.

18 Salesforce, *FY23 Annual Report*, 8.

Customers, often with the help of consultants or system integrators, used Platform to customize implementations of their Salesforce CRM products (e.g., Sales Cloud, Marketing Cloud). Through Platform, customers could add functionality that wasn't included in the standard Salesforce products.

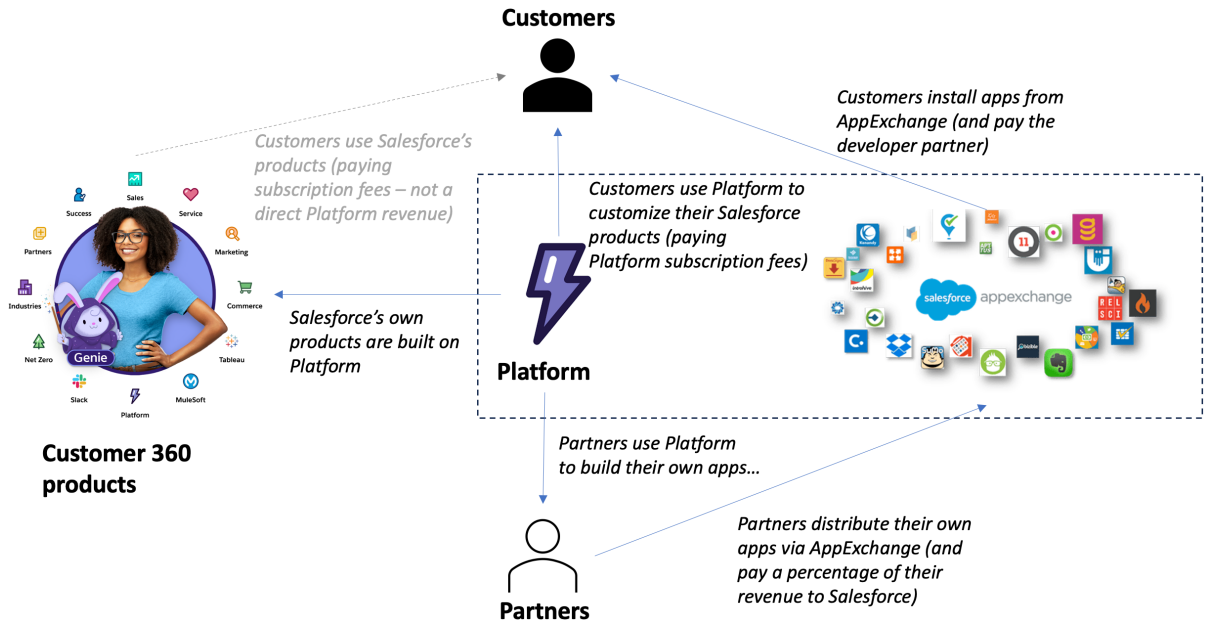
Let's say you're a hospital. One of the simplest customizations our customers make using Platform is this: by default, the customer record type in Salesforce is called an account. [The hospital would] want to call it a patient and add some other objects, for example, about what medications a patient takes. These are pieces of the data model that Salesforce doesn't ship out of the box. So, you make an extension.

PATRICK STOKES, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), PLATFORM, SALESFORCE

Partners such as ISVs[19] used Platform to create their own SaaS solutions (e.g., the DocuSign eSignature app, the Veeva CRM app). In many cases, the partner would offer a solution for download on AppExchange.

In addition, Salesforce's own developers used Platform to build and extend other company products, such as Sales Cloud and Service Cloud. As a result, Platform contributed to Salesforce's revenues both directly, through customer subscription fees and developer royalties, and indirectly, through the enhancement of the company's core products (see figure 1).

Figure 1: A Simplified Illustration of the Salesforce Platform Business Model



Source: The authors.

Starting and Scaling the AppExchange Ecosystem

AppExchange played a key role in Platform's business model; in FY 2023, it contributed roughly $500 million to company revenues. In a 2009 book he co-authored, Salesforce's founder and CEO Marc Benioff wrote, "With AppExchange, we evolved beyond being a technology provider to becoming an enabler of innovative technology. Much of our future rests in this […] ecosystem."[20]

19 The Saleforce.Org General Product Documentation defines ISVs and system integrators in "The Difference between Salesforce System Integrators (SIs) and Independent Software Vendors (ISVs)," Salesforce Help, accessed December 8, 2023, https://help.salesforce.com/s/articleView?id=sfdo.SFDO_Salesforce_SIs_vs_ISVs.htm&type=5.

20 Marc Benioff and Carlye Adler, *Behind the Cloud: The Untold Story of How Salesforce.com Went from Idea to Billion-Dollar Company—and Revolutionized an Industry* (Jossey-Bass, 2009), 126.

In 2023, Salesforce reported that 90 percent of Fortune 500 companies and 91 percent of Salesforce customers used apps from AppExchange.[21] While it wasn't mandatory for partners to distribute their apps via AppExchange, the marketplace grew from 575 listed apps provided by 250 partners in FY 2007 to over 7,000 apps provided by around 4,500 partners in 2023.[22]

AppExchange's growth had been a multiyear journey. Andrew Albert, a former, long-term senior Salesforce executive, described how there were essentially four "chapters" in the AppExchange evolution leading up to 2021.

Chapter 1: Fueling Adoption

It was essential that AppExchange gain well-known partners to use Platform and distribute their apps via AppExchange for the platform to build credibility and external validation in its early phase.

The flagship moment was the partnership between Salesforce and BMC [Software] in 2010. Because the CEO of BMC, the wildly successful market leader in IT services, went on the Dreamforce stage and said, "We are going to build our SaaS offering for a segment of the market on Salesforce." That was the market validation that we were a platform and this was real.

ANDREW ALBERT, CHIEF TECHNOLOGY OFFICER, VIIZR[23]

Chapter 2: Introducing Monetization

Salesforce started Platform as a free offering to encourage adoption. To address increasing resource requirements for building, marketing, and selling Platform, Salesforce turned Platform into a revenue-generating business.

One way for Salesforce to generate revenue was for it to take royalties from partners—a percentage of the revenue they made from selling apps built on Platform. When customers bought an app via AppExchange, Salesforce would collect the revenue, deduct its share, and pass the remainder on to the partner. Partners that sold directly to customers submitted royalties to Salesforce quarterly.

Royalties ranged from 10 percent to 25 percent of app revenues. Apps targeting existing Salesforce customers in the key CRM area were charged less than apps targeting new customers who did not yet subscribe to Salesforce products. And as their app revenues grew, the percentage partners shared fell.

Introducing monetization was tricky because it changed the economic model of partners who had already built on Platform. Salesforce chose a path to revenue that softened the blow for existing partners, increasing gradually the percentage of revenue partners paid to the company.

The monetization of AppExchange turned Salesforce Platform into a revenue-generating business, funding the resources it needed to grow. The monetization approach was supported by two sales pillars: (1) revenue targets, assigned to the partner account managers responsible for the partners' growth, and (2) commissions, paid to the sales representatives responsible for customers who purchased AppExchange apps.

We started comping the customer-facing sales teams. So even people who had no idea their customers were buying these apps got paid. That caused a lot of goodwill internally. Initially, these were reasonably small amounts. But the rest of the Salesforce was like, "This AppExchange thing is unbelievably powerful!" And so we went from this side project to being very visible.

ANDREW ALBERT

21 "What is the AppExchange?," Salesforce AppExchange, Salesforce, accessed November 6, 2023, https://appexchange.salesforce.com/mktcollections/curated/whatisappexchange.

22 From Mark Benioff's introductory letter to stockholders in Salesforce, *Annual Report 2007*, March 9, 2007, on the Salesforce Investor Relations website, https://s23.q4cdn.com/574569502/files/doc_financials/2007/fy07_annual_report.pdf; "What is AppExchange?," accessed September 9, 2023.

23 Andrew Albert became Chief Technology Officer at VIIZR in July 2022. Interviewed in March 2023, he shared his experience from his former role as Senior Vice President, AppExchange at Salesforce, which he left in July 2021.

Chapter 3: Optimizing for Scale

As AppExchange grew, Salesforce needed ways both to efficiently prioritize its thousands of partners and the apps they had built on Platform and to focus limited resources on those partners that were best aligned with Salesforce's strategic goals.

We described ourselves as VCs without money. All we had was our time and experience. We had to figure out who were the right partners, who do I spend my time with.

ANDREW ALBERT, CHIEF TECHNOLOGY OFFICER, VIIZR

To prioritize, the AppExchange team introduced a system to organize partners into four tiers. Partners could earn points in categories important to Salesforce, such as revenue generation, adoption of Platform functionality, and app star ratings.

Revenue is clearly important, but so is technology adoption. We had some partners that were, let's say, 82nd on the revenue list, but with every pilot, every new capability, they jumped in headfirst. Those are early adopters and fun partners. They showcase innovations.

ANDREW ALBERT

With more points, partners could reach higher tiers—and each of the tiers opened up new benefits, including co-marketing with Salesforce (to drive awareness of their apps), access to more advanced technology features and Salesforce experts, and participation in Partner Advisory Boards.[24]

Another factor in prioritization was how well-versed partners were in Salesforce's technologies—measured, for example, by which Trailhead training modules their employees had successfully completed; and to what extent they had participated in partner-focused programs, such as free Partner Learning Camps.[25] The company had found that educated partners were more likely to create reliable apps.

Partners can really screw up things for customers if they steer them in the wrong direction by leaning on the wrong technology, for example. We reward partners for getting educated about our technology and how they can co-sell with Salesforce. Partners can waste a lot of time for our customer account teams and confuse customers if they don't understand the Salesforce pitch.

WOODSON MARTIN, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), APPEXCHANGE, SALESFORCE

Chapter 4: Enabling New Business Models

Over time, Salesforce gained momentum in attracting to Platform as partners established companies, such as Ford Motor Company (in 2021) and Walmart (in 2023), that were not traditionally technology firms but that sought to explore new, revenue-generating digital offerings. Ford, in partnership with Salesforce, launched VIIZR, a SaaS tool that allowed small businesses and trade services (e.g., plumbers, electricians) to manage workers in the field more effectively.[26] Walmart developed the GoLocal and Store Assist apps on Platform and offered them

24 "Partner Advisory Boards," Partner Community, Salesforce, accessed September 9, 2023, https://partners.salesforce.com/pdx/s/learn/article/partner-advisory-boards-MCQCNTVAXEBJHYNHVNZ3ZHYW3QRU?language=en_US.

25 "Partner Learning Camp," Partner Community, Salesforce, accessed December 14, 2023, https://partners.salesforce.com/pdx/s/learn/article/partner-learning-camp-MCUQJPCCSLWRGYRO6N7P5ITTRFZU?language=en_US.

26 Joseph White, "Ford partners with Salesforce to expand software business," Reuters, December 7, 2021, https://www.reuters.com/technology/ford-partners-with-salesforce-expand-software-business-2021-12-07/. Ford reportedly discontinued VIIZR in November 2023 as the offering "met with more competition and limited demand;" see Kirsten Korosec, "Ford shutters company building an app for plumbers, electricians and other trades," TechCrunch, November 2, 2023, https://techcrunch.com/2023/11/02/ford-shutters-company-building-an-app-for-plumbers-electricians-and-other-trades/.

via AppExchange;[27] these apps allowed other retailers to license Walmart's home delivery service for their own stores, speeding up the preparation of customer orders for curbside delivery.

PRIORITIZING PLATFORM RESOURCES

With each new release, the Platform product management team enhanced or added features for Platform, thus allowing third-party developers to build increasingly powerful SaaS applications. For example, with the introduction of large language model capabilities from Salesforce's 2023 launch of Einstein GPT,[28] developers were able to "build [their] own smart assistant" into their SaaS applications.[29] Yet as the number of innovations and product features increased, management needed to prioritize to which features to allocate available resources.

A key question was whether Platform investments that addressed partner or customer requests (and thus generated direct revenue) should be prioritized over internal feature requests.

This is the hard part in running a platform: You want a platform to provide all the coherent building blocks to all the internal applications, but we also want to make money. Striking the right balance between the revenue-generating and the company-supporting aspects [of the platform] is super important. If you don't address that, you're going to end up doing a whole bunch of things that are very targeted to get some more revenue in the short term, and the rest will suffer. Here's the art of it: For the benefit of the company, you don't want to make 20 million from Platform but then lose 40 million in Sales and Service Cloud contracts because you didn't do something they needed.

EMIN GERBA, EXECUTIVE VICE PRESIDENT AND HEAD OF PLATFORM ENGINEERING, SALESFORCE

A second question was how one internal request should be prioritized over another.

Let's take Sales Cloud, a multibillion dollar product, and Healthcare and Life Sciences Cloud, one of our Industries clouds—also a successful product, but not multibillion yet. How do I prioritize one's needs over the other's? The simple way would be to say, you make more money, so I'm going to prioritize you—but that won't work for long. The main way we actually prioritize is with our corporate values.

PATRICK STOKES, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), PLATFORM, SALESFORCE

Salesforce's corporate values were an important mechanism guiding strategic investment priorities during each three-year-horizon long-range planning cycle and an annual planning exercise. The exercise leveraged a management framework that Salesforce CEO and founder Marc Benioff had developed called V2MOM,[30] which was organized around five themes: vision, values, methods (to achieve the vision), obstacles, and measures (to overcome the obstacles). During the V2MOM exercise, each team and individual defined what the themes represented for them. They then linked each theme to the goals of the organization and shared their answers organization-wide.

The first corporate value was trust. For the Platform product management team, this meant that the majority of all resources for Platform development were invested into features that would ensure that Platform remained a

27 Melissa Rapko, "Walmart strikes a deal with Salesforce to sell more of its tech to other retailers," *CNBC*, January 12, 2023, https://www.cnbc.com/2023/01/12/walmart-teams-with-salesforce-to-sell-tech-to-retailers.html.

28 Andrew Cook, "The Definitive Guide to Einstein GPT (Salesforce AI)," Salesforce Ben, October 27, 2023, https://www.salesforceben.com/the-definitive-guide-to-einstein-gpt/.

29 See more on AI-powered Salesforce assistants in "Use the Einstein Platform," part of the Salesforce Einstein Basics "trail"—a guided learning path—on Salesforce Trailhead, accessed December 11, 2023, https://trailhead.salesforce.com/content/learn/modules/get_smart_einstein_feat/use-einstein-platform.

30 Salesforce CEO and founder Marc Benioff developed the V2MOM to provide clarity and alignment on the company's long-term vision and guide decision-making at Salesforce. Marc Benioff, "Create Strategic Company Alignment With a V2MOM," The 360 Blog, May 1, 2020, https://www.salesforce.com/blog/how-to-create-alignment-within-your-company/.

trustworthy platform to build apps on. Other corporate values that helped prioritize investments were customer success and innovation.[31]

We will prioritize trust-related work: In any given moment, north of 50 percent of our resources are [spent] on increasing security, compliance, and accessibility, as an ongoing investment.

PATRICK STOKES, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), PLATFORM, SALESFORCE

Platform product leaders used additional mechanisms for resolving competing investment requests when it wasn't clear which investment would contribute more to a certain corporate value or when investment requirements came up unanticipatedly.

This is where you start to get into human conversations. I say, look, this is my capacity. I can slot you in here. If you want to be up higher, you can go to our chief product officer and ask him to tell us to reprioritize. [But] that's not a good option; you don't want to constantly be bothering the big boss. There's a better option, one that is important for the health of the platform: Give me some of your people. I will put them into the Platform engineering teams, and then we can do what you want because then we will have more capacity.

PATRICK STOKES

Having other product teams provide additional resources that would report temporarily to the Platform engineering team was aimed at avoiding the creation of one-off functionality that was not fit for reuse later on.

If [non-Platform teams] don't get their way, they'll tend to just go build it themselves. And when they do that, they'll do it in an un-Platform way. They'll take the fastest path and whatever they build might not be customizable or extensible by our customers. There's a downstream impact of deviating from the platform.

PATRICK STOKES

If conflicts couldn't be resolved any other way, the general manager, as the P&L owner, made the final decision.

Salesforce is a highly matrixed organization. So the GMs (general managers) own the product management folks, but they don't own the engineers. Then how does the GM truly own the P&L? By having the ultimate decision-making [power] on the priority.

PATRICK STOKES

Engineering the Technology Behind Platform

Platform is the heart of the Salesforce ecosystem because it is the foundational collection of capabilities that allows apps to be built the Salesforce way.

LEO TRAN, DISTINGUISHED ARCHITECT, PLATFORM, SALESFORCE

Like the product management team, Platform's engineering team had to manage serving multiple constituents. The team's foremost responsibility was to quickly provide internal and external developers with the tools and software capabilities to create SaaS apps in the cloud. At the same time, the team also had to make sure that it architected the platform well so the team could maintain it efficiently long-term.

31 "Who We Are," About Us, Salesforce, accessed November 7, 2023, https://www.salesforce.org/about/.

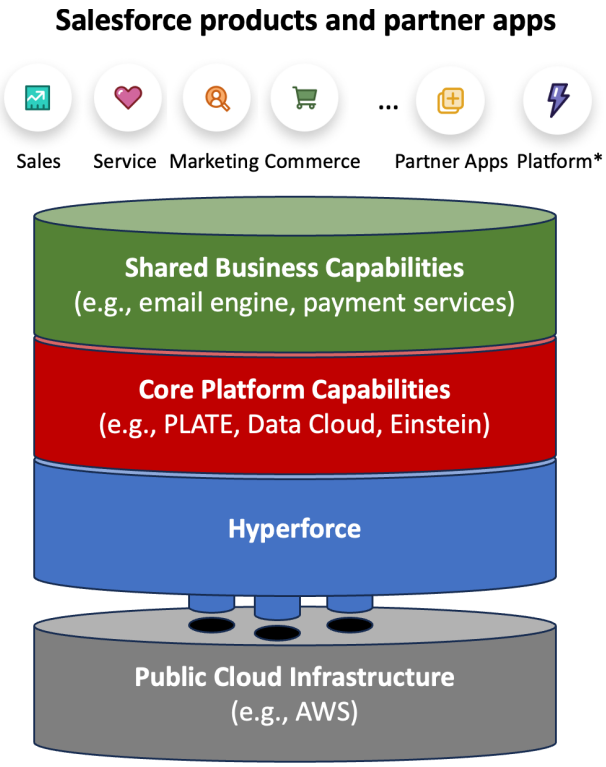
PROVIDING TOOLS TO DEVELOP SAAS APPS

To allow developers to create Platform-based SaaS applications, and to provide a seamless developer experience (DX), the Platform engineering team provided a suite of no-code, low-code, and high-code tools.[32] The tools had evolved over time; initially, in the early 2000s, Salesforce offered a graphical low-code tool that allowed developers to use "clicks, not code" to create Platform-based apps.[33] In 2006, the company added its own Java-like programming language, Apex, to allow developers to build apps with richer business logic.[34] The Platform engineering team was also responsible for developing the software behind the AppExchange site that allowed developers to distribute their apps. See appendix D for an overview of how developers could use these tools to develop a SaaS app.

PROVIDING SOFTWARE CAPABILITIES TO DEVELOP SAAS APPS

To build a SaaS app in the cloud, developers needed various digital capabilities: some cloud infrastructure, such as servers and database management systems; core functionality, to store and edit data in a database; and higher-level business functionality, such as the ability to send email campaigns or to unify customer data from various sources into a single profile. Salesforce had structured these capabilities in what Emin Gerba, the company's executive vice president and head of Platform Engineering, referred to as a "layer cake" architecture (see figure 2).

Figure 2: A Simplified Diagram of Salesforce Platform Architecture



*The Salesforce Platform product exposed the capabilities of the underlying platform for access by customers and partners.

Source: Author adaptation of a diagram from "Hyperforce Info Sheet," Salesforce, accessed December 12, 2023, https://www.salesforce.com/form/platform/hyperforce-datasheet/.

32 The suite included no-code tools such as Object Manager, an interface for managing all objects in a Salesforce environment; low-code tools such as Lightning App Builder, a point-and-click tool for creating the user interface pages; and high-code tools such as extensions (e.g., for debugging) for Microsoft's Visual Studio Code integrated development environment (IDE), which was the de-facto standard for developers building apps on Platform.

33 "Clicks, not code" referred to the easy-to-use declarative programming used by some Salesforce tools that employed blocks of pre-written code packaged into components that could be selected as needed. See a description in "Click, Not Code: The Benefits of Declarative Programming vs. Imperative Programming," Salesforce, accessed December 12, 2023, https://www.salesforce.com/products/platform/best-practices/declarative-programming-vs-imperative-programming/.

34 "Get Started with Apex," Salesforce Trailhead, accessed December 12, 2023, https://trailhead.salesforce.com/content/learn/modules/apex_database/apex_database_intro.

Utilizing Public Cloud Infrastructure

While Salesforce initially operated on its own cloud infrastructure, the company increasingly provided its products on public cloud infrastructures such as Amazon Web Services (AWS).

The decision to start using public cloud infrastructure, rather than do it all ourselves, was applying a true platform strategy, which is to build or leverage best-in-class. If we're going to be a platform provider, we need to aim to be the leader, and we have to provide something which is scalable and extensible. If we can't do that as efficiently in a specific area, then we should use the guys who can and innovate on that.

GAYAN BENEDICT, VICE PRESIDENT OF ARCHITECTURE AND INDUSTRY TRANSFORMATION, ASIA PACIFIC, SALESFORCE AND CO-FOUNDER OF THE SALESFORCE ARCHITECTURE GUILD

Platform apps connected to public cloud infrastructure through Hyperforce, a software layer sitting between the cloud infrastructure provider and higher Platform layers. Hyperforce served as an intermediary to keep the higher layers independent of the underlying infrastructure, "enabling companies to choose between […] AWS, Google Cloud Platform (GCP), and Microsoft Azure."[35]

Providing Core Platform Capabilities

It's a huge challenge: how to create a platform that supports customization and does not let people break things.

PARKER HARRIS, CO-FOUNDER AND CHIEF TECHNOLOGY OFFICER, SALESFORCE

All apps built on Platform needed to use five core capabilities that the Platform engineering team referred to collectively as PLATE: Packaging, Layered Extension, Access Control, Multi-Tenancy, and Entities.

Multitenancy and Access Control: Enabling Many Organizations To Use the Same Cloud Application

Platform housed data and functionality for multiple, distinct user groups at the same time, known as multitenancy. Platform's engineers had to make sure that users from one customer organization could engage with their group's data and customizations while being completely isolated from any other customer organization's data and customizations. To achieve this without maintaining costly separate databases or virtual machines for each customer, Platform associated a customer-specific unique ID, called a Salesforce Organization ID—or Org ID—with each row of data a customer organization added to the shared central database.[36]

Entities: Enabling Developers to Add Custom Data Without Breaking the Platform

A core capability used by any app built on Platform was inserting, updating, and retrieving data about entities such as client accounts or products, which Salesforce referred to as objects. Instead of accessing the database directly, however, all apps had to send requests for retrieving or changing data via a data abstraction model.[37] This approach ensured that any request to the database included a customer's Org ID.

The temptation is to just reach down all the way to the data, because that's the fastest way to do it. But that creates spaghetti code.

EMIN GERBA, EXECUTIVE VICE PRESIDENT AND HEAD OF PLATFORM ENGINEERING, SALESFORCE

35 "Salesforce Hyperforce Will Unlock A New Level of Global Compliance," Plative, accessed November 10, 2023, https://plative.com/salesforce-hyperforce/.

36 For a detailed explanation of Salesforce Platform's approach to implementing multitenancy, see Ian Varley, "L33T M10y," Medium, January 24, 2018, https://medium.com/salesforce-engineering/l33t-m10y-f04f38127b82.

37 A data abstraction model (or layer) is a mechanism for providing access to data in a database indirectly via APIs rather than directly. Platform used this approach to separate the underlying data storage and retrieval mechanisms from the rest of the application.

Platform's data abstraction model and its proprietary database querying language called SOQL (Salesforce Object Query Language) allowed Platform to stay independent from whatever database management system (e.g., Oracle Database) Salesforce used on the infrastructural layer. As a result, Salesforce could freely switch between data storage systems if needed.[38] In addition, having developers use SOQL allowed Platform to optimize database queries for improved performance.

Platform used metadata to allow partners and customers to create custom objects (e.g., a "Property" object for a real estate brokerage), or even custom code to be triggered by a precipitating event.[39] Where data represented facts—such as the name or address of a client account—metadata described information about that data, and it could be leveraged to influence the behavior of Platform for a specific organization. For example, if a developer created a custom object for "Property," all the information about the custom object—that it was called Property, that it included an Address field—was stored, along with the customer's unique Org ID, in a database table for all custom objects. This way, whenever a user of a specific organization logged-in to Salesforce, Platform could "see" in the table for custom objects that this user should have access to a specific custom object or that certain code should be executed when the user's activity triggered it.

Layered Extension: Enabling Coexistence of Customizations by Different Platform Stakeholders

Whenever the engineering team updated the core platform—they made three releases per year—Salesforce needed to ensure that the customizations and additions by customers or ISVs would remain functional. To enable this, Salesforce designed a capability it called layered extension. The core platform was the inner layer, ISV applications were the next layer, and customer customizations were the outer layer. Each layer could be changed without affecting the other layers.

Our partners can modify Service Cloud and make it work for the healthcare industry by changing it and adding capabilities. And then a customer can further modify it for their own use. And then we as Salesforce can change the base platform without breaking anything. We're all doing that constantly. We have to ensure that the entire system remains trusted, that it remains scalable, secure, available, and that it's bug free. That's very, very hard.

PARKER HARRIS, CO-FOUNDER AND CHIEF TECHNOLOGY OFFICER, SALESFORCE

Layered extension prevented naming collisions. It permitted, for example, an ISV to create a custom object called "Property," and the engineering team to add a standard object also called "Property" in a new release of a core Salesforce product, with the custom and standard Property objects coexisting and the ISV app, with its custom Property object, continuing to work. To achieve this, Platform introduced namespace prefixes and specific suffixes into the names of its custom objects; a custom "Property" object developed by an ISV might actually be referred to as ISVxyOrg__Property__c within Platform. Another aspect of layered extension was that Platform enforced a certain order of code execution,[40] with customer-developed code executed before partner-developed code, and partner-developed code executed before Salesforce's own code. Managing these challenges added complexity to Platform.

What happens when we save a record in Salesforce is it actually goes through twenty steps.

LEO TRAN, DISTINGUISHED ARCHITECT, PLATFORM, SALESFORCE

38 Having its own query language also allowed Platform to support more and different data types—such as "email" and "URL," along with format validation—than standard SQL databases would provide. In most SQL databases, an email address or URL would be conveyed as a text (variable character) data type that did not come with any kind of inherent format validation.

39 Ian Varley, "Metadata: Software The Way You Want It," Salesforce Engineering Blog, accessed November 8, 2023, https://engineering.salesforce.com/metadata-software-the-way-you-want-it-2367b179558d/.

40 "Salesforce Platform Order of Execution Overview," Salesforce Architects, October 28, 2022, https://architect.salesforce.com/1/asset/immutable/s/e6cf2ac/assets/images/Salesforce-Order-Of-Execution-Diagram.png.

Salesforce also required developers of any app deployed on AppExchange to provide additional testing functionality that would test at least seventy-five percent of the entire code of the app. Before Salesforce updated Platform, it could automatically run these tests for apps to either rework its update or to notify developers of potential issues.[41]

Packaging: Enabling Developers to Distribute Their Apps

Platform also had to provide ways for developers to distribute their apps via AppExchange. Salesforce called this concept packaging, as it involved wrapping all the additions and customizations an ISV offered (e.g., custom objects, Apex code for workflows) into a container—the package—that could be "unboxed" when a user bought the ISV app.

Providing Shared Business Capabilities

The top layer of the layer cake platform architecture encompassed all the standard business functionalities (e.g., actions around sending marketing campaign emails, approval processes) that were available to ISVs and customers to reuse and customize. These shared business capabilities were not developed by the Platform engineering team, but by engineering teams responsible for specific Salesforce products (e.g., Salesforce Marketing Cloud). To distribute platform capabilities across multiple engineering teams, some coordination was required to avoid inconsistencies that would affect ease-of-use for developers.

To developers, it's all Platform, even though different engineering teams focus on different areas. And that's been problematic in terms of engineering org design. We're trying to solve this problem by designing APIs with consistent, coherent interfaces—having more governance on, for example, what APIs look like—and consistent, coherent API documentation.

LEO TRAN, DISTINGUISHED ARCHITECT, PLATFORM, SALESFORCE

Balancing the Needs for a Healthy Architecture and Time to Market

Indeed, making Platform easy for developers to use was a key objective for the Platform Engineering team.

The first step in getting adoption is building a platform that is easy to use.

PARKER HARRIS, CO-FOUNDER AND CHIEF TECHNOLOGY OFFICER, SALESFORCE

Platform engineers viewed as an important factor influencing ease-of-use that all provided platform functionality followed a uniform underlying logic. Once developers understood the underlying logic, they would be able to use different functionality with less effort.

What many platforms miss is that unified, coherent view. Oftentimes, you have a series of microservices, [each with its] own metadata. But what we hear back from customers is, "What I really like is if I do customizations on one object in Sales Cloud but I also have Service Cloud, that I don't need two specialized developers, because the metadata model is coherent." It takes a lot of deliberate intent to maintain that coherence. We have to invest in it when we would rather invest in some short-term improvements. But in the long run, the attractiveness of a platform is influenced by those investments.

EMIN GERBA, EXECUTIVE VICE PRESIDENT AND HEAD OF PLATFORM ENGINEERING, SALESFORCE

Maintaining that overall coherence was a challenge not only because different engineering teams worked on shared capabilities but also due to the important need to develop functionality at high speed. Designing new functionality too quickly risked taking shortcuts that would create inconsistencies in Platform's architecture.

41 "Get Started with Apex Unit Tests," Apex Testing, Salesforce Trailhead, accessed December 15, 2023, https://trailhead.salesforce.com/content/learn/modules/apex_testing/apex_testing_intro?trailmix_creator_id=strailhead&trailmix_slug=journey-to-salesforce-2.

A cross-product Architecture Review Board (ARB) consisting of architects from each product engineering team decided whether a certain piece of functionality should be (1) a reusable platform or business capability or (2) a unique, one-off feature within a specific product.

Building platform capabilities versus functionality in an app is the everyday tension. I've gone through this over the past few months with generative AI. An intelligent chatbot—Generative AI—isn't an app. It needs to be a reusable capability because I'm going to use this chatbot across [both] Sales and Service Cloud. But even I, as an architect, agree that time to market is super important. Having the killer app first is super important. And so part of our job is figuring out how we iterate towards the desirable architecture.

LEO TRAN, DISTINGUISHED ARCHITECT, PLATFORM, SALESFORCE

In situations where the most desirable architecture could not be achieved immediately, defining a roadmap to the desired architecture up front was considered key.

We're consciously saying "we're taking this shortcut," which sounds obvious but isn't necessarily done in the rush of things. When we make this conscious decision, we need to determine the path to the right architecture. We need to say, "okay, we are doing this, but this is what we ought to be doing, and here's how we get from where we are today to where we want to go."

EMIN GERBA, EXECUTIVE VICE PRESIDENT AND HEAD OF PLATFORM ENGINEERING, SALESFORCE

Shortcuts could not be avoided completely, even though they eventually necessitated time-consuming clean-up work later on.

Now, do we always go back and do it? No. We don't. When you start accumulating these things and you see problems on the horizon, then you have to have these long-running refactoring efforts.

EMIN GERBA

The PLATE capabilities helped define a set of "no discussion" rules that all development teams needed to adhere to—without shortcuts. For example, regardless of time-to-market pressure, a new app would not be allowed to directly access the database by circumventing the use of the data abstraction model APIs.

The value of PLATE was to assert what were important capabilities of Salesforce's architecture to reuse and align to. […] PLATE does evolve, but we don't want multiple and inconsistent implementations of PLATE capabilities.

LEO TRAN

SERVING BOTH EXTERNAL AND INTERNAL PLATFORM STAKEHOLDERS

Most Platform capabilities were needed by both external developers (customers and partners) and internal Salesforce developers. After all, all of Salesforce's own products, such as Sales Cloud and Service Cloud, also needed to leverage shared capabilities such as supporting multiple tenants and altering data entities.

There were some capabilities, though, that engineers needed to create specifically for external developers, such as the packaging capability used for app distribution via AppExchange. Having external developers as stakeholders also influenced the versioning of capabilities; for example, all APIs had a minimum support lifetime of eighteen months, so that external developers would have sufficient time to update their apps before APIs they used were decommissioned. And because it was impossible to foresee the types of apps external developers would build on top of Platform, its functionality always had to be written without any particular assumptions about the context in which it would be used.

If we did not provide external developers access to Platform, there would be less complexity. Once an API exists, the amount of permutation is astronomical. People do all kinds of things.

EMIN GERBA, EXECUTIVE VICE PRESIDENT AND HEAD OF PLATFORM ENGINEERING, SALESFORCE

For example, Platform engineers could not have foreseen that one company would build an entire postal management system on top of Platform, and another one a core banking system. First using functionality in Salesforce's own products, before exposing the functionality externally via APIs, helped engineering teams develop an understanding of its possible uses.

If we need to get something out for Sales Cloud, we know how it's going to be used. That is the first step. And then we have this larger step of "now we're going to open this capability to everyone and release it into the wild! Let's see what people do!"

EMIN GERBA

Building and Nurturing a Community of Developers

Attracting and retaining external developers to build on Platform was crucial to the success of its business model.

If I want to convince the CIOs of the world's largest companies to bet on the Salesforce Platform, what better demonstration could there be than entrepreneurs who bet their whole business on the Salesforce Platform? So we actively cultivated an ecosystem of entrepreneurs to build dedicated applications.

WOODSON MARTIN, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), APPEXCHANGE, SALESFORCE

To cultivate a strong community, Salesforce established a Technical Audience Relations team, composed of roughly forty people within the company's marketing organization. The technical audiences the team targeted included developers, admins, architects, and designers. Within the team, the Developer Relations group helped create content—such as articles, videos, audio recordings, presentations, and sample apps—to enable developers to generate apps, maintained various channels to deliver content to developers, and managed a community of developers.

PROVIDING RESOURCES TO HELP DEVELOPERS BUILD APPS

Content delivery channels for the Developer Relations team included a blog dedicated to developers;[42] the Salesforce Developers YouTube channel—founded in 2008 and boasting 126,000 subscribers and 2,400 videos by December 2023;[43] the Salesforce Developers Podcast;[44] and several conferences,[45] including the annual Dreamforce conference with 40,000 attendees in 2023 and a dedicated annual developer conference, TrailblazerDX.[46] Salesforce's Developer Portal (developer.salesforce.com) was the hub for all developer resources, including API documentation. Salesforce's learning website Trailhead offered free training courses that Platform developers could take to earn certifications.[47]

42 Salesforce Developers Blog, Salesforce Developers Portal, accessed December 13, 2023, https://developer.salesforce.com/blogs.

43 Salesforce Developers channel, YouTube, accessed December 13, 2023, https://www.youtube.com/SalesforceDevelopers.

44 Salesforce Developers Podcast, Salesforce Developers Portal, accessed December 13, 2023, https://developer.salesforce.com/podcast.

45 Salesforce Developers Events, Salesforce Developers Portal, accessed December 13, 2023, https://developer.salesforce.com/events.

46 Sumeet Wadhwani, "Dreamforce 2023: On AI, CRM, Data, Partnerships, San Francisco and More," *Spiceworks*, September 15, 2023, https://www.spiceworks.com/tech/big-data/articles/dreamforce-2023-highlights/.

47 Salesforce Platform Basics, Salesforce Trailhead, accessed December 13, 2023, https://trailhead.salesforce.com/content/learn/modules/starting_force_com.

SCALING AND SUSTAINING THE DEVELOPER COMMUNITY

Another key element to developer relations was to grow and maintain a community of dedicated developers.

That has been part of the secret sauce, creating that community where people are not only inspired by the technology, but by that sense of community, belonging, feeling people wanting to help each other. That is unique about the Salesforce community and it is a key pillar of developer relations.

CHRISTOPHE COENRAETS, SENIOR VICE PRESIDENT, TECHNICAL AUDIENCE RELATIONS, SALESFORCE

Anyone could become part of the Salesforce Trailblazer community by simply signing up on the Trailhead portal.[48] Joining the community allowed Trailblazers to connect with each other and interact—by asking or answering Salesforce-related questions—in forums.

Beyond helping developers build apps on Platform, the Developer Relations team helped developers build a professional career with their Salesforce-related skills. For example, one section of the landing page of the Trailhead portal called "Kickstart Your Career" highlighted what a career for a developer specializing in Salesforce looked like—for example, a "$120,000 average salary" with "52,270 jobs [posted] in the last 12 months" based on US data—and how to get started with free training material and other resources.[49]

Initially, the Developer Relations team had started out as evangelists.

People in the community got tired of that model in the entire industry of, "hey, look at me. I'm that celebrity. I'm going to tell you the right way of doing things." An evangelist is a rock star. "This is my content; I'm going to go deliver it all over the world, and that's what makes me great." That was the mentality in the early days of evangelism. And now our mission is not to be rock stars but to inspire people and enable them.

CHRISTOPHE COENRAETS

But by 2023, members of the team aimed to act as developer advocates instead.

Being a developer advocate is a humbler positioning. You do presentations, you give keynotes, but there's a collaborative vibe. It's not just "you listening to me," it's [also] me listening to you by creating a feedback loop. That's one of the key changes.

CHRISTOPHE COENRAETS

As just one example, at the "True to the Core" session held at the Dreamforce 2023 conference, the company's engineering leadership, including the CTO, answered questions from the developer community—about adding functionality, improving usability and performance, the product roadmap, and more—and responded to suggestions for improving the developer experience.[50]

Becoming developers' trusted advisors was a key reason to emphasize the developer advocate role over the evangelism approach.

When we describe a technical solution for them, they trust that we are not trying to sell them a product because that's better for our bottom line. They trust that this is the best technical solution for them.

CHRISTOPHE COENRAETS

48 Trailblazer Community, Salesforce Trailhead, accessed December 13, 2023, https://trailhead.salesforce.com/en/trailblazercommunity.

49 Salesforce Trailhead, accessed December 13, 2023, https://trailhead.salesforce.com/.

50 Session attended by case authors. A recording of the session is available online: "True to the Core Customer Forum," Salesforce+, accessed January 15, 2024, https://www.salesforce.com/plus/experience/dreamforce_2023/series/best_of_dreamforce_at_dreamforce_2023/episode/episode-s1e9.

Another reason to shift away from the evangelism approach was the need to scale.

To me, the key thing with a small team like mine is scale, scale, scale. I've been in teams before where you have a lot more people, but all they do is travel the world. It doesn't scale that well.

CHRISTOPHE COENRAETS, SENIOR VICE PRESIDENT, TECHNICAL AUDIENCE RELATIONS, SALESFORCE

To help scale, Salesforce also supported the developer community in its efforts to self-organize many events. The Salesforce Ben website, a go-to resource around everything Salesforce-related, listed over twenty community-organized events in 2023.[51] The Developer Relations team provided materials that community developers could use to educate other developers.

The Developer Relations team has the concept of "presentation in a box." We say, "hey, member of the community, you want to deliver a session on that topic? Here's a package— You have the slide presentation, the demos, everything you need."

CHRISTOPHE COENRAETS

EXTENDING THE SCOPE OF THE PLATFORM COMMUNITY

In 2018 and 2021, Salesforce created teams of five and twelve people respectively to foster communities of architects and designers—roles that were highly relevant to the adoption of Platform.

Architects drive a lot of important decisions relating to the usage of our products and the health of a solution. Architects are involved in deciding whether a customer buys Salesforce products and how they should be implemented.

ZAYNE TURNER, SENIOR DIRECTOR, ARCHITECT RELATIONS, SALESFORCE

To better support the needs of software, solution, and enterprise architects, the Architect Relations team developed the Salesforce Well-Architected Framework, a collection of resources including whitepapers and tools that helped architects build Trusted (secure, compliant, and reliable), Easy, and Adaptable applications on Platform.[52]

Similarly, the Designer Relations team sought to help user experience (UX) designers for Salesforce-based applications design the best user experiences by supplying resources such as design guidelines and certifications. A key tool was the Lightning Design System,[53] a framework of patterns and user interface components providing a unified language and consistent look and feel when designing apps for the Salesforce ecosystem.

Developers don't necessarily come to the table with the human-centered design training or background. Our Salesforce Lightning Design System is a way of scaling best practices in human centered design so all makers and builders can create experiences our customer love. It basically says, "okay, developer, every button you put on a page, in this context needs to look like this and behave like this."

ADAM DOTI, VICE PRESIDENT AND PRINCIPAL DESIGN ARCHITECT, SALESFORCE

Linking to other roles beyond developers helped Salesforce become integrated at multiple touchpoints with companies building apps based on Platform:

51 Events, Salesforce Ben, accessed September 9, 2023, https://www.salesforceben.com/salesforce-events-2023/.

52 Salesforce Architects, last updated July 2023, https://architect.salesforce.com/well-architected/overview.

53 The Lightning Design System was created by the Salesforce Design System team for developers to ensure an accessible and scalable product experience. Lightning Design System, accessed December 13, 2023, https://www.lightningdesignsystem.com/.

In my mind a healthy ecosystem makes you too big to fail. A key to health in an ecosystem is understanding who the stakeholder groups are and should be.

ZAYNE TURNER, SENIOR DIRECTOR, ARCHITECT RELATIONS, SALESFORCE

At the same time, supporting a new role required a share of limited resources, and the Designer Relations team had to navigate prioritization challenges:

My team is striving for equitable representation and access to the levers that other teams and roles have to be successful and support their audiences' growth. If there is a Trailblazer ecosystem event or a campaign, don't forget the UX designers! To build the designer community, everybody in the ecosystem needs to share the responsibility to advocate, elevate, and celebrate the designer audience.

ADAM DOTI, VICE PRESIDENT AND PRINCIPAL DESIGN ARCHITECT, SALESFORCE

Marketing AppExchange to Partners and Customers

In early 2023, about fifty people were working in AppExchange Marketing, promoting AppExchange to both customers who might purchase apps from the marketplace and partners who could provide apps.

It's a different type of product marketing, because we are thinking not just about the customer experience, but also working to optimize the marketplace experience and programs for partners. It's a balance.

MONICA GIROLAMI, VICE PRESIDENT, APPEXCHANGE PRODUCT MARKETING AND COMMUNITY (FORMER), SALESFORCE[54]

HELPING PARTNERS MARKET THEIR APPS TO CUSTOMERS

The AppExchange marketing team helped partners better position their apps towards Salesforce's customers.

A lot of our emerging partners do not have marketing teams (yet); they are developers and engineers. So we provide partner programs and training to help these emerging entrepreneurs think through the resources, and value-driven messaging and go-to-market programs to best support their listings. You will hear a lot of partners asking for "more leads." And so we offer programs to help [partners] build lead generation programs with Salesforce and stand alone to help them achieve success.

MONICA GIROLAMI

The AppExchange Marketing Program (AMP) was a pay-to-play vehicle that allowed partners to promote their apps to generate more leads. (See appendix E for a list of sample AMP offerings and pricing.) For example, the AMP offered partners options to pay for co-branded videos to showcase an app, to advertise the app on the AppExchange home page, to create app-specific Trailhead learning modules, and to present their app to Salesforce employees to increase company awareness of the app.[55]

I previously led North American marketing at NewVoiceMedia [now Vonage], a top Salesforce AppExchange partner. A large part of my marketing strategy was focused on joint go-to-market customer events with Salesforce, and driving awareness with Salesforce sellers on the benefits of joint selling, our joint competitive differentiation, and how and when to recommend our joint solution.

MONICA GIROLAMI

54 Monica Girolami left her role at Salesforce in July 2023.

55 Partner Co-Marketing Program, Salesforce AppExchange, accessed December 13, 2023, https://appexchange.salesforce.com/mktcollections/curated/appexchange-marketing-program-amp.

ATTRACTING THE RIGHT PARTNERS

Partner marketing started with recruiting new partners who would complement Salesforce's CRM products with new functionality. The marketing team was proactively looking for partners to fill white space in Salesforce's offerings, especially to showcase the capabilities of new technologies, such as generative AI.

Why have an ecosystem of third-party software applications around your product? To complete the product. Salesforce might have 80 percent or 90 percent of the capability in the platform, but there's always something special that people need. Partners can fill that gap and develop a business and grow it. And our customers can benefit from something that's ready to install and fit for purpose.

WOODSON MARTIN, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), APPEXCHANGE, SALESFORCE

Once new partners had been recruited, the marketing team also facilitated their onboarding. Publishing an app on AppExchange involved signing up online to the Partner Community program[56] and completing documentation of the app. Documenting the app involved providing an app name, description, and pricing, as well as compliance-related information, and a business plan that ideally demonstrated a revenue opportunity that justified Salesforce's investment in the partner. After the partner signed a commercial agreement with Salesforce, every app had to pass a security review, which took on average two weeks.

COORDINATING WITH OTHER PRODUCT ENGINEERING AND MARKETING TEAMS

One challenge the partner teams needed to address at times arose when Salesforce product engineering teams extended the functionality of their product—or when Salesforce acquired a company that brought in functionality—that directly or indirectly superseded functionality that was also addressed by a partner's solution.

Our product teams continue to engineer, we continue to acquire new technology […]. As these events occur, it does require us to change our AppExchange marketplace positioning and it can also change the market opportunity for a Salesforce ISV partner in a positive or negative way. When these events occur, our whole Partner organization is working closely with our ISV partners to manage this change and our joint success outcomes.

MONICA GIROLAMI, VICE PRESIDENT, APPEXCHANGE PRODUCT MARKETING AND COMMUNITY (FORMER), SALESFORCE

To help product marketing teams from other divisions (such as Sales Cloud or Service Cloud) recognize and promote the value of AppExchange to customers, the AppExchange marketing team created collections of apps for each Salesforce product.

Many of our customers know the value of AppExchange ISVs because they couldn't extend Salesforce to meet their needs without them—it's the power of the Salesforce platform! Internally, not all product marketers understand the value and how AppExchange ISVs support and extend their [Sales, Service, Marketing, etc.] Cloud. For this reason, we have teams that focus on building collections of ISVs' apps by Cloud, use case, and industry on the marketplace, and others that partner with our Cloud product marketers to enrich their Cloud solution value story with ISV app collections.

MONICA GIROLAMI

In 2023, Salesforce moved the AppExchange marketing team from the Platform product organization to the company's central marketing organization to improve its integration with Salesforce's broader marketing efforts.

56 Salesforce Partner Community, accessed December 13, 2023, https://partners.salesforce.com/pdx/s/.

Selling Platform to Salesforce Customers

The market for Platform-as-a-Service was very competitive: Amazon offered AWS; Microsoft, Azure; and Google, its Google Cloud platform. Competing platforms were more generic than Salesforce's Platform, which was built for use cases around CRM functionality. The focus on CRM functionality could help make the development of these types of apps faster and easier than when using a generic PaaS. It also meant that other types of SaaS applications—those outside the CRM spectrum—were not as easy to build on top of Platform.

It's easier to build apps around people, around business logic, like CRM apps on Platform. It's much harder to do social media apps, for example. You don't really want to build a Twitter or a Facebook on Platform.

LEO TRAN, DISTINGUISHED ARCHITECT, PLATFORM, SALESFORCE

This specialization had an impact on the types of sales Salesforce was most successful in closing.

Our sweet spot is customers who have made substantial investments in Salesforce clouds and thus, Platform is a natural extension. It makes sense to build a new loyalty app on the Salesforce platform, for example, because you can leverage the data living in your Salesforce CRM. If you are going into a net new environment, competing with the other platform providers in a generic development shop, that's a harder [area to] compete.

ROBERT WICKHAM, VICE PRESIDENT AND GENERAL MANAGER, TABLEAU ASIA PACIFIC, SALESFORCE[57]

SELLING A PLATFORM VERSUS SELLING AN APPLICATION

Selling the Salesforce Platform is very challenging, because we're not known widely as a platform company. We are viewed as an applications company.

ROBERT WICKHAM

At Salesforce, prime sellers were the salespeople who sold core applications like Sales Cloud or Service Cloud to customers. Prime sellers typically held the key contacts at customer companies. A complication in selling Platform was that it involved a different kind of conversation than most prime sales people were used to.

Our prime selling teams know how to engage customers around applications, which are pre-built capabilities that solve for a particular use case. How do you make salespeople more productive? That's a Sales Cloud use case. But Platform is like a box of Lego bricks. You can build anything with it, so you start with a blank canvas, and say, "what do you want to build?" That's a different conversation, one that many traditional Salesforce sellers are not familiar with.

ROBERT WICKHAM

Selling Platform also involved talking to different types of people than those prime sellers usually did. Unlike sales meetings for core CRM products, sales meetings for Platform involved people interested in more technical aspects such as architecture and security. These meetings engaged people from different departments.

The personas are also different. The primary personas for Salesforce are heads of sales service and marketing. […] With Platform, it's typically the developer and infrastructure teams.

ROBERT WICKHAM

57 When the authors interviewed him in June 2023, Robert Wickham shared experiences from when he held the role of Area Vice President, Platform and Emerging Technologies at Salesforce in 2019.

Thus, Platform required its own specialized salespeople who would act as co-prime sellers. These individuals had to make two sales pitches: the first internally, to a prime seller; and the second externally, to the customer.

A prime salesperson only has so much mindshare capacity and our portfolio of sellable products has become relatively large. Specialist sellers compete for that mindshare internally. For many prime sellers, Platform is not in their comfort zone, so they would gravitate to other Salesforce products which they are more comfortable with.

ROBERT WICKHAM, VICE PRESIDENT AND GENERAL MANAGER, TABLEAU ASIA PACIFIC, SALESFORCE

The required skillset for successful Platform salespeople—a deep understanding of Platform's technical capabilities—was different from that of other salespeople.

Trust is vital between a specialist and prime seller. For a specialist, a large part of this trust is built on credibility, which comes from subject matter expertise. It's akin to being a language translator—imagine being in a meeting in Japan and your translator admits to still ramping [up] and not being able to translate everything. They would immediately lose credibility with you. If, however, you had an amazing experience with that translator, chances are you would recommend them to all your colleagues.

ROBERT WICKHAM

These skills were not always easy to find or develop, though.

If you're hiring classic salespeople, they most likely have strong selling skills but not the specific expertise in platforms. You hope they pick it up as quickly as possible. Then, you get into that Japanese translator challenge, because they can't insert themselves from day one. On the other hand, subject matter experts may not possess the strong selling skills needed to progress and close opportunities. The unicorns you are looking for in a specialist sales person possess both.

ROBERT WICKHAM

LOOKING AHEAD: OPPORTUNITIES AND CHALLENGES

Building and maintaining an ecosystem connecting multiple stakeholders had plenty of internal and external challenges, as Patrick Stokes, the former general manager for Platform, summarized:

By far the biggest difference between Platform and our other products is thinking through how do I build a relationship with the field sales force, how do I enable them on our products, and how do I sell Platform to customers. And then figuring out how to build relationships with all of the internal folks that need something from you. It's company politics. They'll escalate and create all sorts of challenges for you in prioritizing your work.

PATRICK STOKES, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), PLATFORM, SALESFORCE

With a successful ecosystem around Salesforce Platform in place, a key question for management was how to continue the growth trajectory.

Salesforce was increasingly helping new, nontraditional partners build their own SaaS digital offerings, including winning Walmart in 2023 as an ISV on the AppExchange. At the same time, some financial analysts were worried about big customers like Veeva Systems leaving the platform.[58] Founded in 2007 by a former Salesforce executive, Veeva offered its SaaS application with CRM-like functionality for companies in the life sciences industry on

58 Jordan Novet, "Salesforce faces an underappreciated risk: Platform customers jumping ship like Veeva," *CNBC*, March 4, 2023, https://www.cnbc.com/2023/03/04/salesforce-faces-prospect-of-platform-customers-leaving-after-veeva.html.

AppExchange. But in 2023, Veeva announced it would stop building its products on Salesforce Platform in 2025 and instead use its own platform. One financial markets content service wrote:

Eventually, switching to its own platform will help with margins down the line. However, the bigger opportunity is that it will allow VEEV to move into some adjacent industries […] that it previously wasn't allowed to compete in on the CRM side due to its agreement with Salesforce. Moving away from Salesforce's technology is also a risk. Salesforce will be allowed to compete for life science customers, once the agreement winds down, if it wants to focus on this vertical.[59]

The AppExchange team considered evolving its revenue model for partners.

AppExchange's economic model makes sense if you have many small companies. But many of our partners have grown beyond that. The fixed percentage rate of the royalty model makes it very difficult for some of our largest partners to increase their gross margins, because if they raise their prices, we just take more of it. We make one-off agreements to solve for that today, but it's a point of high friction. We need an economic model evolution. Consumption pricing is interesting.

WOODSON MARTIN, EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER (FORMER), APPEXCHANGE, SALESFORCE

Amazon's cloud platform, for example, had a pricing model based on consumption, in which developers paid for the data or compute their app actually used. Salesforce had to find a model that would work for its customers and partners and made sense economically for the company.

We need to think about consumption at a business measure level. For example, DocuSign wants to pay per envelope, not per CPU hour. So how do we build a model where, for a wide variety of diverse application types, there are key business measures that matter to both the customer and the partner and that can be matched to a cost structure and a margin model on our end?

WOODSON MARTIN

One promising option for growth was extending into industries that had previously been left to ISVs. The challenge was to resist the temptation to build various industry *products* and instead continue applying *platform* thinking.

We take requirements that are specific to an industry and think about how we translate that into a platform capability—not a hard-coded piece of functionality. For example, a capability of tracking lifetime events—birth, marriage, divorce, death—is interesting from a healthcare and a wealth management perspective. Underneath it's a common platform capability. That takes a lot of skill and patience. It takes longer to build out a capability and make it platform-centric and metadata-driven than to solve a specific use case.

PARKER HARRIS, CO-FOUNDER AND CHIEF TECHNOLOGY OFFICER, SALESFORCE

Salesforce also continued to add new functionality to Platform; in 2023, the focus was generative AI. At its annual Dreamforce 2023 conference, Salesforce announced it was rebranding Platform as "Einstein 1 Platform" as it added shared platform capabilities and tools for creating prompts for large language models while masking the data and checking for the appropriateness of the generated answers.[60] (See appendix F for an overview of the new Einstein features in Platform).

59 Geoffrey Seiler, "Veeva Moving Away from Salesforce Creates an Opportunity," Seeking Alpha, January 31, 2023, https://seekingalpha.com/article/4573811-veeva-moving-away-from-salesforce-creates-an-opportunity.

60 "Salesforce Announces the New Einstein 1 Platform – Driving Productivity and Trusted Customer Experiences Powered by Data, AI, and CRM," News & Insights, Salesforce, September 12, 2023, https://www.salesforce.com/news/press-releases/2023/09/12/salesforce-platform-news-dreamforce/.

We have been figuring out how to bring the power of AI to our customers with Salesforce values like trust. How do we make sure that the AI is based on ethics, that customers are not giving their data to competitors because the AI will learn from their data? How do we protect our customers from hallucinations—the AI says something very confidently and someone works with a customer and does the wrong thing, says the wrong thing, sends the wrong email, proposes the wrong treatment in healthcare, whatever it is. How do we help our customers know how to trust this AI? It's a whole new world we're going into. We are going as fast as we can, guided by our values.

PARKER HARRIS, CO-FOUNDER AND CHIEF TECHNOLOGY OFFICER, SALESFORCE

Platform had created growth opportunities for its customers and partners; at the same time, it held many opportunities to grow itself. Leadership was balancing the risk of straying too far from Salesforce's core products by turning Platform into an even more generic application development platform.

First and foremost, we are a CRM company, not a platform company. Our view of Platform is that it is related to our business, CRM. If you think about a platform generically, as a technical platform to build whatever applications you want, you can lose sight of why you are building a platform, if you're not careful. Salesforce has been successful with its Platform because we remained laser-focused—a new Platform capability for what kind of customer and use case—so we don't get wrapped up in technology for technology's sake.

PARKER HARRIS

APPENDICES

Appendix A: Selected Salesforce Financials

	2015	2016	2017	2018	2019	2020	2021	2022	2023	CAGR*
Total revenues (in millions)	$ 5,374	$ 6,667	$ 8,437	$ 10,540	$ 13,282	$ 17,098	$ 21,252	$ 26,492	$ 31,352	25%
Net income (in millions)	$ -263	$ -47	$ 323	$ 360	$ 1,110	$ 126	$ 4,072	$ 1,444	$ 208	
Return on sales	-5%	-1%	4%	3%	8%	1%	19%	5%	1%	
Total Subscription and Support revenue (in millions)	$ 5,014	$ 6,206	$ 7,799	$ 9,766	$ 12,413	$ 16,043	$ 19,976	$ 24,657	$ 29,021	25%
% of total revenues	93%	93%	92%	93%	93%	94%	94%	93%	93%	
Platform & Other revenues (in millions)	$ 745	$ 1,035	$ 1,433	$ 1,913	$ 2,854	$ 2,787	$ 3,324	$ 4,509	$ 5,967	30%
% of total revenues	14%	16%	17%	18%	21%	16%	16%	17%	19%	
% of total subscription and support revenues	15%	17%	18%	20%	23%	17%	17%	18%	21%	

* CAGR = Compound annual growth rate

Source: Data compiled from Salesforce annual reports by the authors.

Appendix B: Customer 360 Product Catalog



Figure B.1: A depiction of Salesforce products forming part of the Customer 360 product catalog.

Source: "Customer 360," Salesforce, accessed August 20, 2023, https://www.salesforce.com/products/.

Appendix C: Interviewees by Function

The authors conducted interviews for this case study between November 2, 2022 and August 2, 2023.

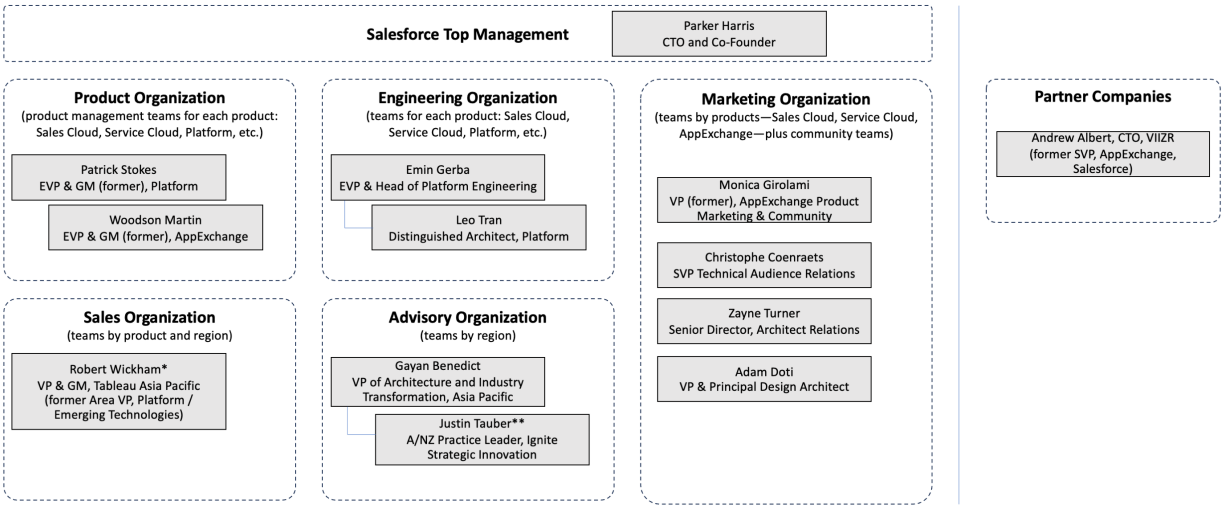


Figure C.1: An organizational diagram of individuals interviewed by the authors for this case study showing their role at Salesforce.

** The authors interviewed Robert Wickham to provide insights from his earlier role within the Sales organization at Salesforce, although at the time of the interview he was serving in a role at Tableau.*

*** The authors interviewed Justin Tauber for this research but did not quote him in this case study.*

Source: Compiled by the authors.

Appendix D: Overview of How a Developer Would Build an App on Platform

Platform tools resided within Setup in Salesforce Cloud products. In simplified terms, a developer employed the tools in the following way to create a new app. The developer would start by clicking the Setup cogwheel icon in a product to access the Object Manager. In the Object Manager, the developer would build the app by adding Custom Objects to it using a setup wizard. Objects represented entities about which the app would store data. Salesforce came with many standard objects, such as Account, which stored data about customer accounts; and Case, which stored data about incidents, such as complaints by a customer. The developer could create custom objects and custom fields to tailor the app for a specific purpose; for example, for an app for use by a property brokerage company, the developer might create custom objects such as "Property" and "Broker," adding custom fields to the object—such as "Address" and "Availability status" for a property object—using the Object Manager.

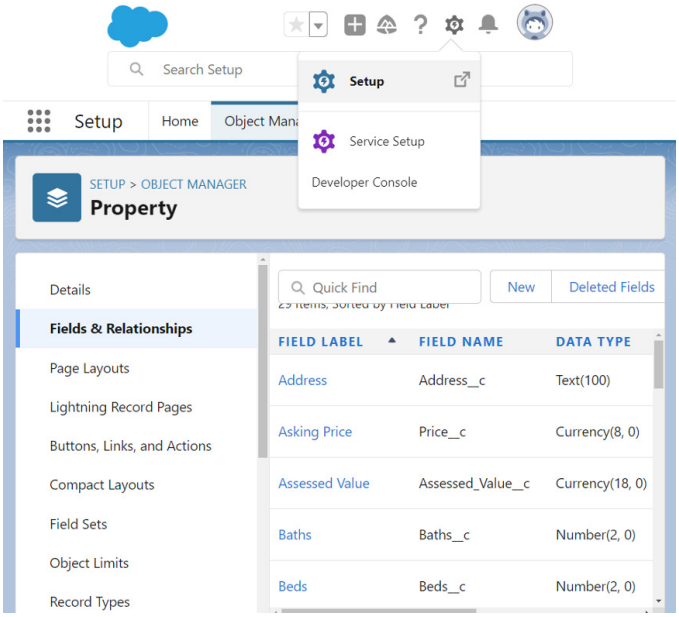


Figure D.1: Screenshot of the Object Manager, displaying a custom object for Property.

Source: Author's screenshot from DreamHouse, a sample app Salesforce provided for developers.

The developer could then use the Lightning App Builder to graphically design a user interface by which a customer would create or edit a record—such as a new property, in the case of a property brokerage company—using elements such as text fields and action buttons and determining their layout and appearance.

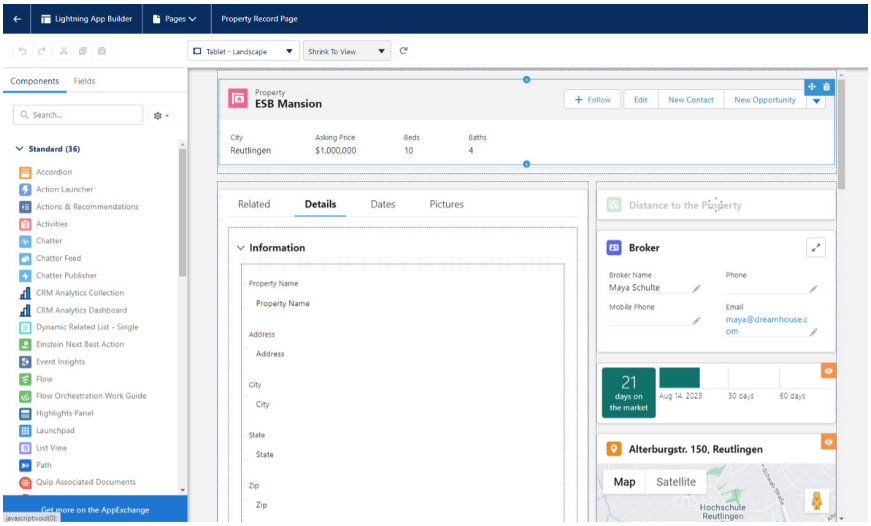


Figure D.2: Screenshot of the Lightning App Builder used to define the user interface of a page.

Source: Author's screenshot from DreamHouse, a sample app Salesforce provided for developers.

The developer additionally could use a tool called Flow Builder to add workflows to execute a process—such as to require that a supervisor first approve a broker's assignment to a property before the assignment would be completed.



Figure D.3: Screenshot of the Flow Builder tool used to create workflows for an app.

Source: Author's screenshot from DreamHouse, a sample app Salesforce provided for developers.

Users experienced the app through tabs for each object—such as a tab for properties, another for brokers, and so forth. Users could easily execute actions, such as to add new properties and brokers or to change the status of a property, in order to conduct business efficiently.

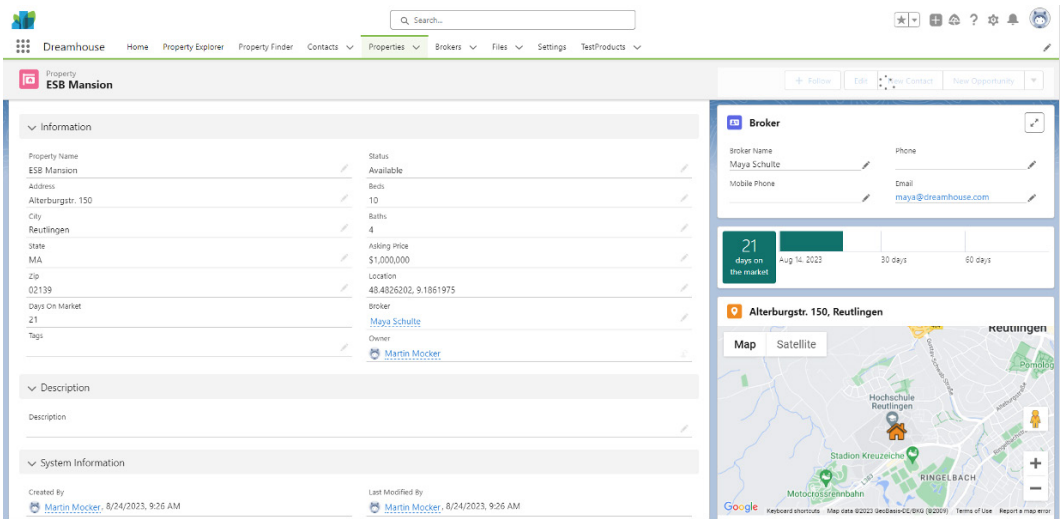


Figure D.4: Screenshot of the user interface by which a user would experience an app.

Source: Author's screenshot from DreamHouse, a sample app Salesforce provided for developers.

Appendix E: Examples of Promotional Offerings from the AppExchange Marketing Program (AMP) to Partners

Category	Offering	Pricing (USD)
Showcase your product	Marketing team co-creates different types of co-branded video to explain the key features of a partner's app	$10,000–$17,500
Promote your AppExchange listing	Add a chat feature to an AppExchange listing that allows potential customers to interact with the partner providing the app	$15,000 annually
	Display app listing in the "Sponsored" section of the AppExchange homepage	$15,000 per quarter
	Co-branded display banner ads	$20,000 per quarter
	Sponsor keywords for the AppExchange search	$7,500 per quarter
Increase solution awareness	Create a Trailhead module for a partner app	$50,000
	Purchase an advertorial in the quarterly digital AppExchange magazine	$20,000
Educate Salesforce employees	Run a "Demo Jam" to showcase an app to Salesforce Solution Engineers	$7,500

Source: Data compiled by authors from Partner Co-Marketing Program, Salesforce AppExchange, accessed August 31, 2023, https://appexchange.salesforce.com/mktcollections/curated/appexchange-marketing-program-amp.

Appendix F: The Salesforce Marketecture[61]

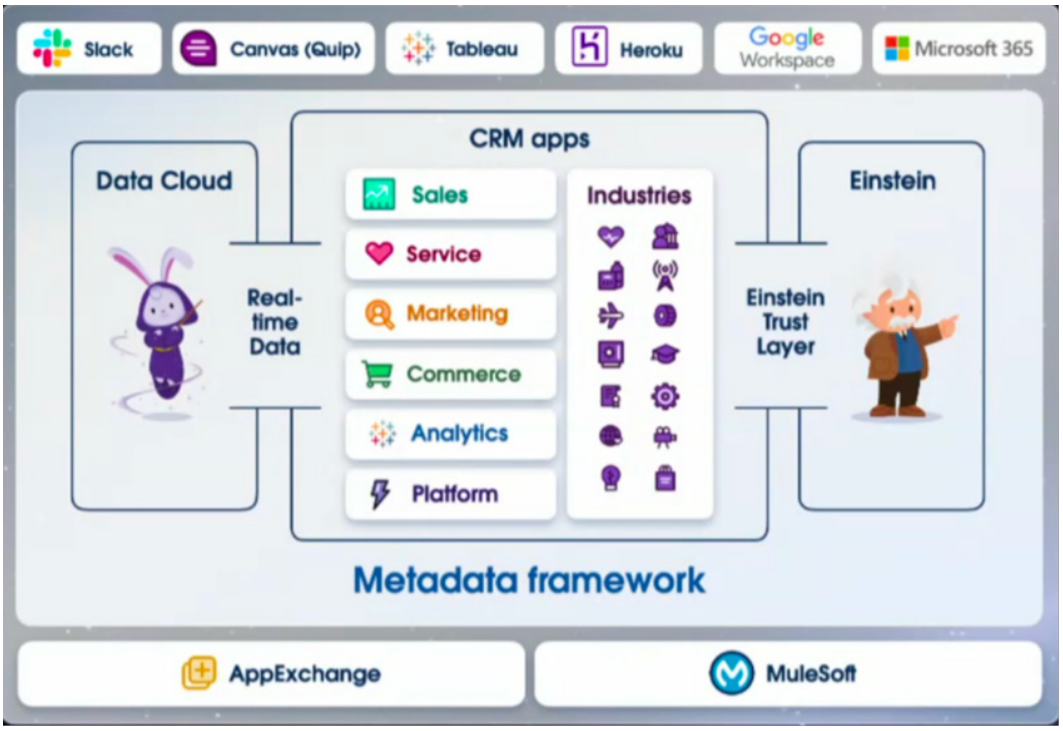


Figure F.1: The Salesforce Marketecture, as presented at Dreamforce 2023.

Source: Screenshot from a Dreamforce 2023 keynote.

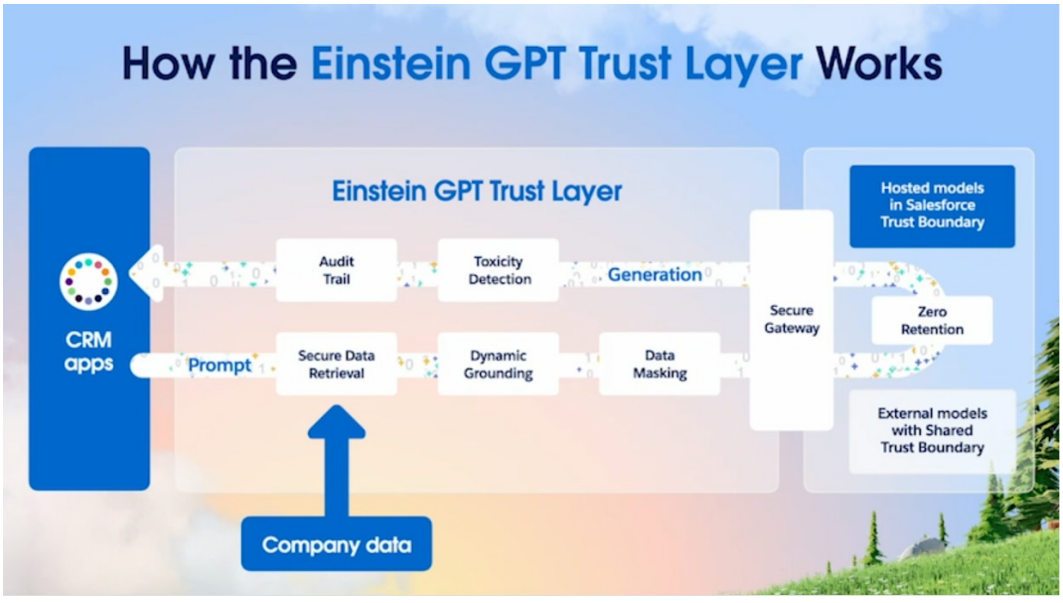


Figure F.2: An Einstein Trust Layer schematic.

Source: "Salesforce Announces AI Cloud: How Salesforce is bringing trusted generative AI to the enterprise," News & Insights, Salesforce, video, accessed December 13, 2023, https://www.salesforce.com/news/stories/video/explaining-the-einstein-gpt-trust-layer/.

61 A marketecture is an architectural depiction for marketing purposes. Salesforce frequently updated its marketecture to highlight new features.

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